

ATM Architecture Design

Software architecture for a simulated ATM session, covering layered responsibilities, main components, session flow, and error handling.



Presentation Roadmap

01  ATM Scope and Goals

02  Layered Software Structure

03  Architecture and Main Flows

04  State Machine and Failures

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ATM Scope and Goals

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04 State Machine and Failures

Supported ATM Functions

🎯 Functional Boundary & Design Intent

The system is limited to a closed session workflow that begins with card insertion and ends with card return. The core design goal is to create a clean, teachable ATM software structure that separates responsibilities, keeps transaction control predictable, and makes exception handling explicit, rather than modeling a full banking platform.

⚙️ Core Supported Functions

1 PIN Entry & Authentication
Supports up to three attempts for user verification.

2 Balance Inquiry
Check current account balance securely.

3 Cash Withdrawal
Dispense cash based on user request and availability.

4 Cash Deposit
Accept cash deposits into the user's account.

5 Receipt Printing
Provide physical transaction records upon request.

🛡️ Exception Handling & Architecture

⚠️ Explicit Exception Scenarios

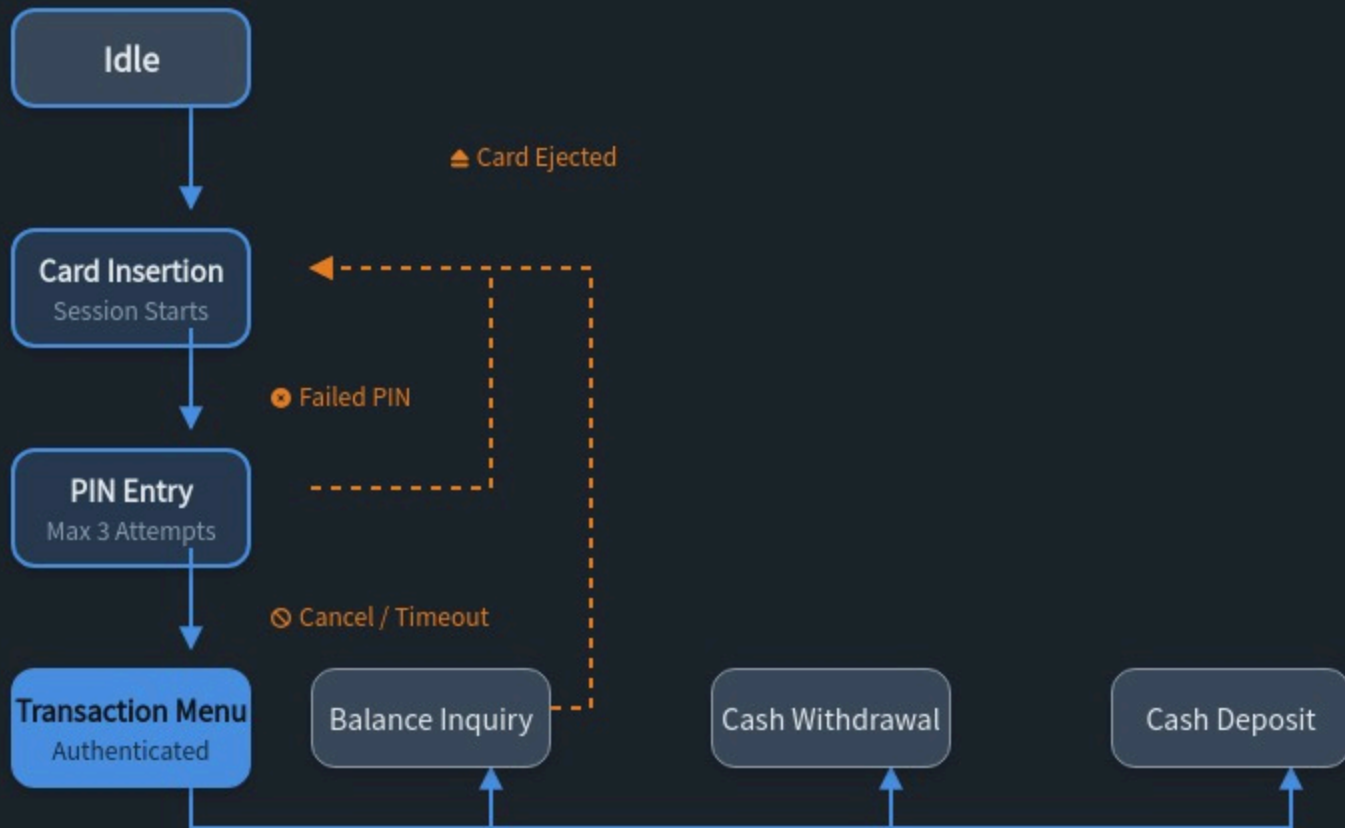
- **Cancel:** User initiates termination of the current session.
- **Timeout:** System terminates session due to user inactivity.
- **Error Closure:** Graceful shutdown on hardware/network failures.

🏗️ Architectural Perspectives

- **User-Flow Perspective:** The user interface remains strictly focused on interaction, ensuring a seamless experience.
- **Component-Responsibility Perspective:** Transaction control is kept predictable with clear separation of duties.

■ Design Goals and Constraints

ATM Session Lifecycle



Key Design Constraints

- **Controlled Session**
Functions are not independent screens. They form a single, controlled session where the card acts as the session token.
- **PIN Verification Gate**
PIN entry gates access to all financial operations. The transaction menu is only exposed after successful authentication.
- **Active Actor**
The authenticated customer is the active actor throughout the session lifecycle.
- **Post-Transaction Output**
Receipt printing is treated as a post-transaction output function, not a separate business process.

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Layered Software Structure

02 Layered Software Structure

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▪ User Interaction Layer



User Interaction

- Responsible for prompts and menus
- Handles message display to users
- Ensures clear separation from backend logic



Transaction Control

- Manages session logic
- Responsible for decision making
- Coordinates between UI and backend services



Device Abstraction

- Handles cash dispenser operations
- Manages card reader interactions
- Controls printer for receipts



Bank Communication

- Performs account verification
- Handles balance requests
- Processes posting requests

Transaction and Device Layers



User Interaction Layer Overview

The user interaction layer contains the visible and input-facing parts of the ATM. Its primary responsibility is to communicate with the user in a way that is consistent, simple, and state-aware, ensuring a seamless transaction experience.

↔ Inputs and Outputs

↪ Visible Parts (Outputs)

- Screen displays (welcome, menus, balances)
- Card reader prompts
- Receipt display logic & success/failure messages

→ Captured Actions (Inputs)

- Keypad entries (PIN digits)
- Menu selections & cancel requests
- Inactivity-related timeouts (session timer)

⚠ Core Architectural Rule

NO Business Decisions

This layer strictly handles presentation and input capture.

What it does NOT do:

- ✗ Decide whether a PIN is correct
- ✗ Determine if cash is available
- ✗ Approve or reject transactions

→ Instead, it forwards user actions to the transaction layer.

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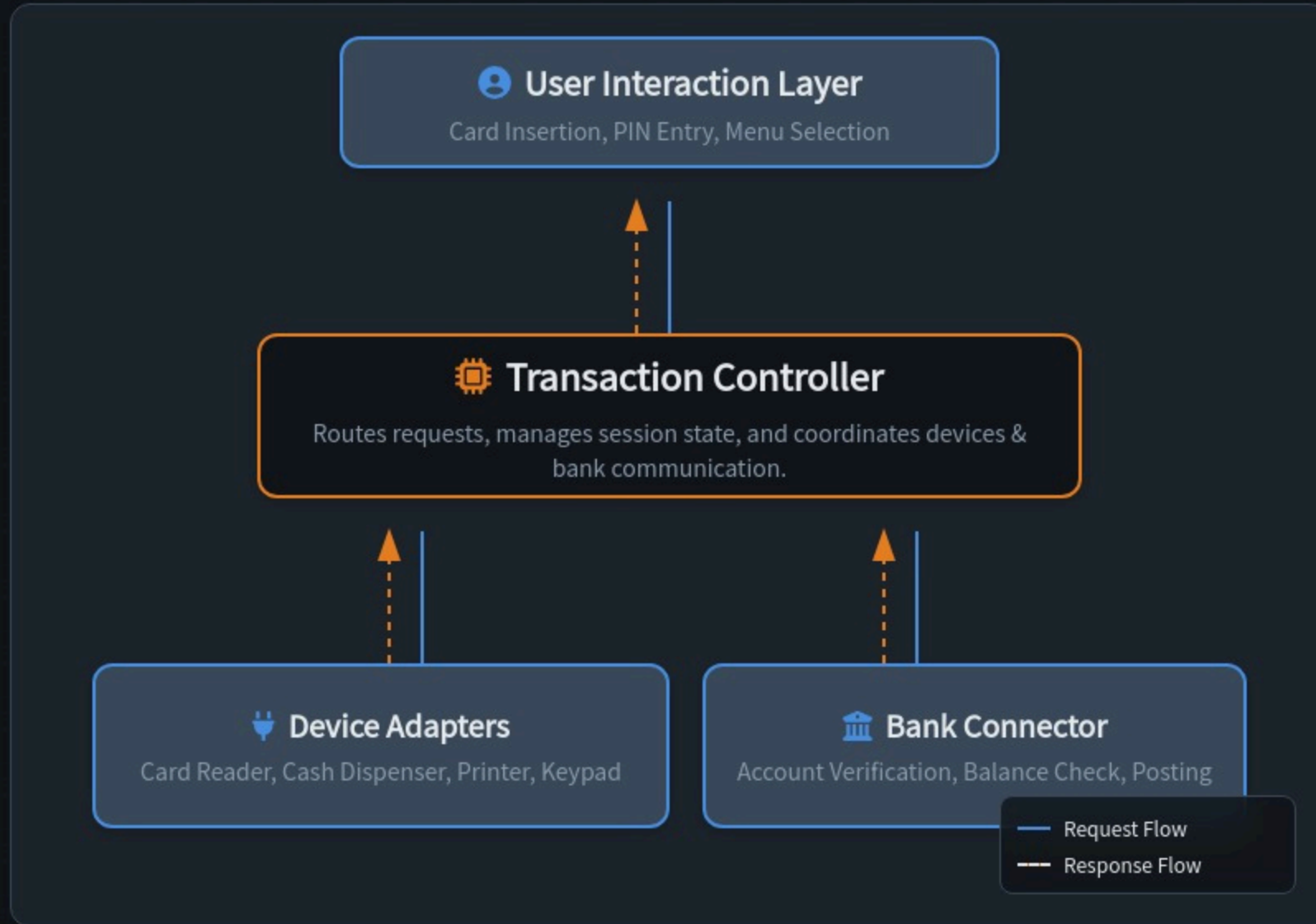
Architecture and Main Flows

06 Auth, Flows, and Errors

05 Transaction Controller and Adapters

06 Auth, Flows, and Errors

■ ATM Layer Interaction



☰ Transaction Flows

- **Session Initiation**

Card insertion triggers the session. PIN verification acts as the primary gatekeeper for access.

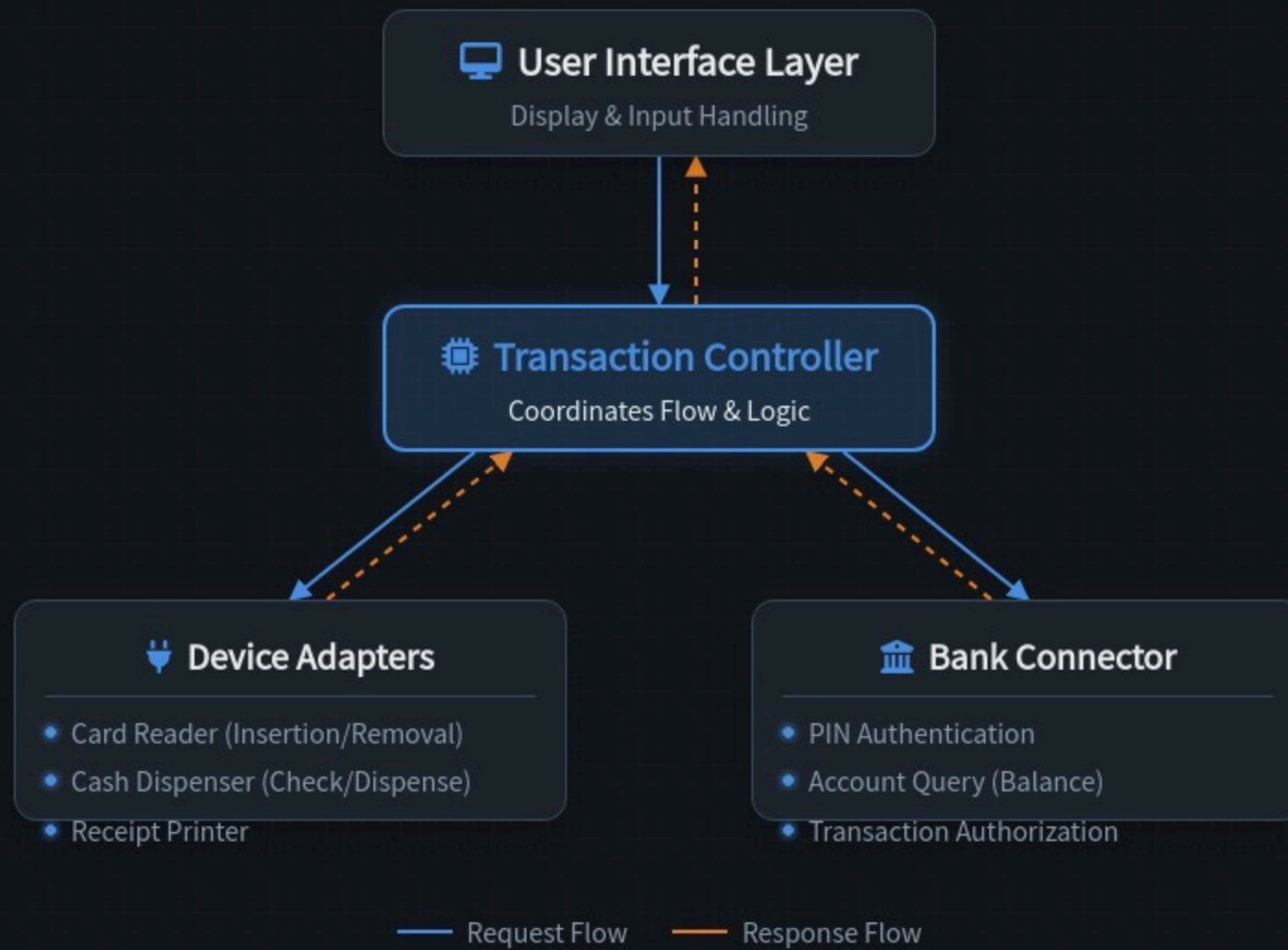
- **Transaction Routing**

The controller routes user selections to specific transactions. Results dictate next steps: print receipt, show menu, or terminate.

➔ Simplified Deposit Path

1. Deposit Request Initiated
2. Account Posting via Bank
3. Receipt Printing

Core Transaction Paths



1 Session Activation

Card insertion begins the session and activates the controller, which then requests PIN entry through the interface.

2 Transaction Menu

If the PIN is accepted, the controller opens the transaction menu and waits for the user to choose balance inquiry, withdrawal, deposit, or cancel.

3 Specific Flows

Balance Inquiry: Controller sends account query via bank connector, forwards balance to interface.

Withdrawal: Controller asks bank for authorization, checks ATM cash availability before committing.

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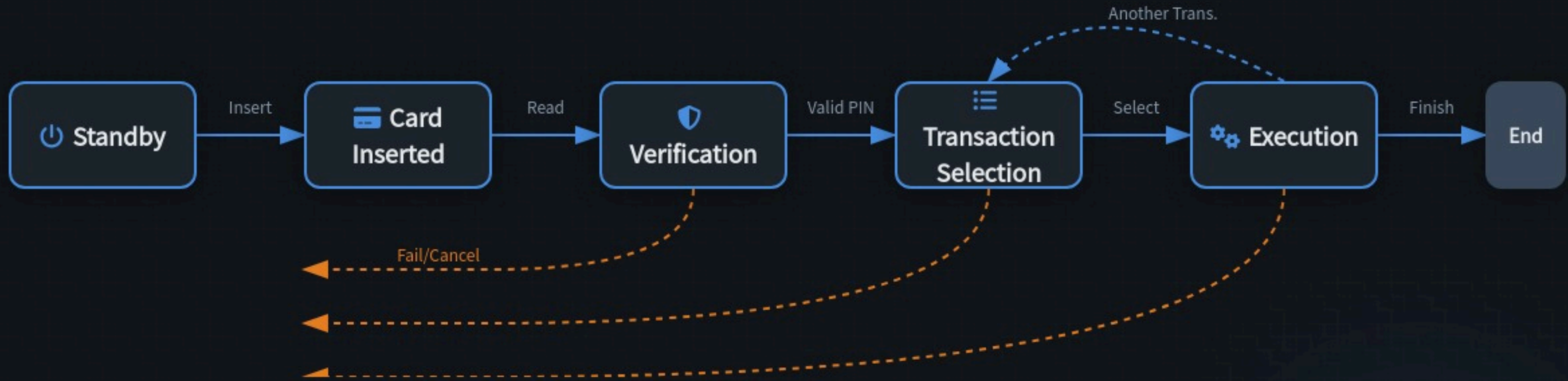
State Machine and Failures

02 State Machine Overview

03 Transaction Paths

04 State Machine and Failures

■ Session State Machine



⚠ Failure Outcomes Summary

Incorrect PIN

Prompt retry; increment counter.

3 Failed PIN Attempts

Card retained or ejected; session terminated.

User Cancellation

Immediate card return; session ends.

User Inactivity Timeout

Session aborted; card returned to user.

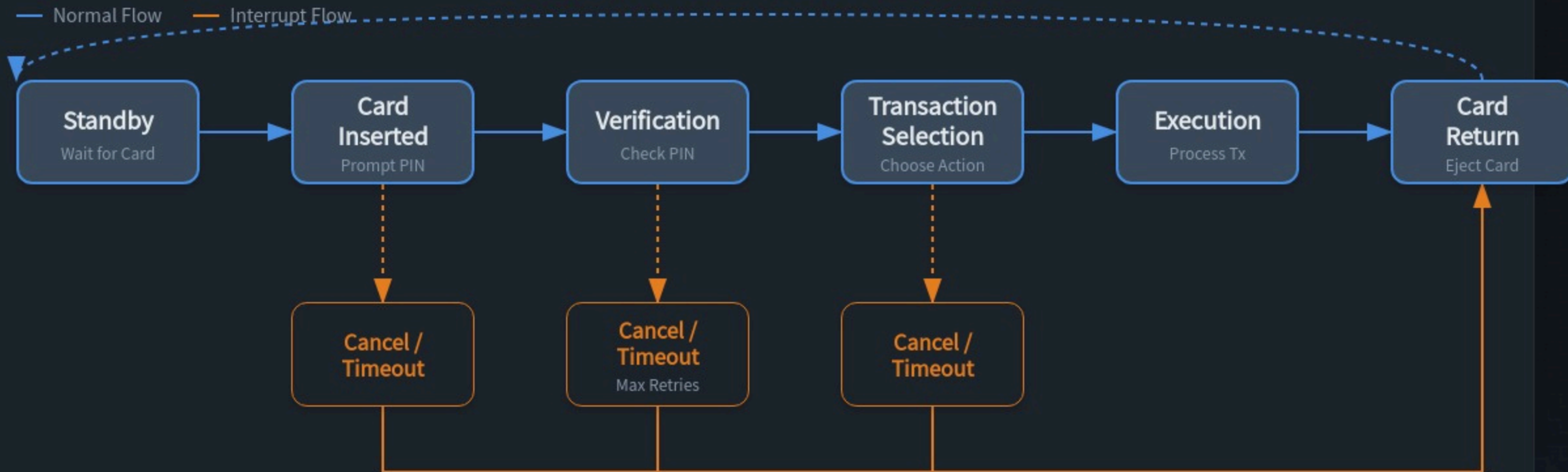
Insufficient ATM Cash

Transaction rejected; prompt new amount.

Bank Connection Failure

Transaction aborted; error message displayed.

■ Error Handling Outcomes



✖ User Cancellation

If the user selects cancel during PIN entry, verification, or transaction selection, the session is immediately interrupted and routes to the Card Return state.

⌚ Inactivity Timeout

A timeout due to user inactivity at any prompt (Card Inserted, Verification, Transaction Selection) triggers an automatic abort, transitioning to Card Return.

🛡 Verification Failure

The Verification state manages retry counts. If authentication fails repeatedly and reaches the maximum allowed attempts, the session aborts to Card Return.

Design Summary

- **Incorrect PIN:** Retry within verification state, attempt count reduced. After three failed attempts, session terminates and card is returned to prevent unlimited retries.
- **User Cancellation:** Immediately stops current action and returns card, regardless of whether during PIN entry or transaction selection.
- **Inactivity Timeout:** ATM closes session, returns card, and returns to standby state.
- **Insufficient Cash:** Withdrawal request fails without dispensing money. Controller rejects transaction, returning user to menu or ending session.

Thank you for reviewing the ATM Software Architecture Design.